
Adaptive Control of a Quadrotor UAV with Uncertain Mass, Inertia, and Control Allocation Matrix

SC617 Project Report

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Date: April 2026

1 Part 1: Adaptive Control Design on SE(3)

1.1 Quadrotor Dynamics Model

The quadrotor is modelled as a rigid body with configuration on $\text{SE}(3) = \mathbb{R}^3 \times \text{SO}(3)$. Let $\mathcal{I} = \{x_i, y_i, z_i\}$ denote the inertial frame and $\mathcal{B} = \{x_b, y_b, z_b\}$ the body-fixed frame attached at the centre of mass. The equations of motion are:

$$\dot{\mathbf{p}} = \mathbf{v} \quad (1)$$

$$m\dot{\mathbf{v}} = T\mathbf{R}\mathbf{z}_i - mg\mathbf{z}_i \quad (2)$$

$$\dot{\mathbf{R}} = \mathbf{R}\hat{\boldsymbol{\Omega}} \quad (3)$$

$$\mathbf{J}\dot{\boldsymbol{\Omega}} = \boldsymbol{\tau} - \boldsymbol{\Omega} \times \mathbf{J}\boldsymbol{\Omega} \quad (4)$$

where $\mathbf{p} \in \mathbb{R}^3$ is inertial position, $\mathbf{v} \in \mathbb{R}^3$ is inertial velocity, $\mathbf{R} \in \text{SO}(3)$ is the rotation from body to inertial frame, $\boldsymbol{\Omega} \in \mathbb{R}^3$ is the body-frame angular velocity, $T \in \mathbb{R}$ is total thrust, $\boldsymbol{\tau} \in \mathbb{R}^3$ is the net body-frame torque, $m \in \mathbb{R}^+$ is mass, $\mathbf{J}$ is the symmetric positive definite inertia matrix, g is gravitational acceleration, and $\mathbf{z}_i = [0, 0, 1]^\top$.

The *hat map* $\hat{\cdot} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ is defined by $\hat{\mathbf{a}}\mathbf{b} = \mathbf{a} \times \mathbf{b}$ for all $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$. Its inverse is the *vee map* $(\cdot)^\vee : \mathfrak{so}(3) \rightarrow \mathbb{R}^3$.

Uncertain parameters and linear-in-parameter form. Both m and $\mathbf{J}$ are unknown. For a diagonal inertia matrix $\mathbf{J} = \text{diag}(\theta_1, \theta_2, \theta_3)$, the rotational dynamics admit:

$$\mathbf{J}\dot{\boldsymbol{\Omega}} + \boldsymbol{\Omega} \times \mathbf{J}\boldsymbol{\Omega} = \boldsymbol{\Psi}_o(\boldsymbol{\Omega}, \dot{\boldsymbol{\Omega}}) \boldsymbol{\theta} \quad (5)$$

where $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^\top$ and the *attitude regressor* $\boldsymbol{\Psi}_o \in \mathbb{R}^{3 \times 3}$ is (see Appendix A):

$$\boldsymbol{\Psi}_o(\boldsymbol{\Omega}, \boldsymbol{\alpha}) = \begin{bmatrix} \alpha_x & -\Omega_z\Omega_y & \Omega_y\Omega_z \\ \Omega_z\Omega_x & \alpha_y & -\Omega_x\Omega_z \\ -\Omega_y\Omega_x & \Omega_x\Omega_y & \alpha_z \end{bmatrix} \quad (6)$$

1.2 Geometric Attitude Error Functions

Working intrinsically on $\text{SO}(3)$ avoids the representation singularities of Euler angles and the unwinding phenomenon of unit quaternions [8].

Definition 1.1 (Geometric Attitude Errors). *Given desired attitude $\mathbf{R}_d(t) \in \text{SO}(3)$ and actual attitude $\mathbf{R}(t) \in \text{SO}(3)$:*

$$\mathbf{e}_R = \frac{1}{2} (\mathbf{R}_d^\top \mathbf{R} - \mathbf{R}^\top \mathbf{R}_d)^\vee \in \mathbb{R}^3 \quad (7)$$

$$\Psi(\mathbf{R}, \mathbf{R}_d) = \frac{1}{2} \text{tr}(\mathbf{I} - \mathbf{R}_d^\top \mathbf{R}) \quad (8)$$

The angular velocity error is $\mathbf{e}_\Omega = \boldsymbol{\Omega} - \mathbf{R}^\top \mathbf{R}_d \boldsymbol{\Omega}_d$ where $\boldsymbol{\Omega}_d = (\mathbf{R}_d^\top \dot{\mathbf{R}}_d)^\vee$.

The function Ψ is positive definite about $\mathbf{R} = \mathbf{R}_d$ with range $[0, 2]$. Its critical points where $\mathbf{e}_R = \mathbf{0}$ are $\mathbf{R} = \mathbf{R}_d$ (minimum, $\Psi = 0$) and a set of measure zero at $\Psi = 2$ (antipodal configurations). This enables almost-global stability results.

1.3 Hierarchical Control Structure

The cascade structure of (1)–(4) motivates a two-loop hierarchy. The *outer loop* (position controller) generates a desired thrust direction and magnitude from position tracking requirements. The *inner loop* (attitude controller) drives the actual attitude to track that desired orientation.

The fundamental insight — borrowed from Pliego-Jiménez [3] and translated here into rotation-matrix language — is to introduce a *coupling term* in the reference angular velocity $\boldsymbol{\Omega}_r$ that links the translational error $\mathbf{s}_p$ to the attitude dynamics. This coupling enables exact algebraic cancellation of the translational-rotational cross-term in the Lyapunov derivative, yielding a single positive-definite Lyapunov function for the complete system without requiring cascade stability arguments.

1.4 Outer Loop: Position Adaptive Controller

1.4.1 Error Variables and Composite Signal

Define position and velocity tracking errors:

$$\tilde{\mathbf{p}} = \mathbf{p}_d - \mathbf{p}, \quad \tilde{\mathbf{v}} = \mathbf{v}_d - \mathbf{v} \quad (9)$$

The reference velocity and composite translational error:

$$\mathbf{v}_r = \mathbf{v}_d + k_p \tilde{\mathbf{p}} \quad (10)$$

$$\mathbf{s}_p = \mathbf{v}_r - \mathbf{v} = \tilde{\mathbf{v}} + k_p \tilde{\mathbf{p}} \quad (11)$$

with $k_p > 0$. Differentiating and substituting the plant (2):

$$m\dot{\mathbf{s}}_p = m\boldsymbol{\psi}_p - \mathbf{f} - T(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i \quad (12)$$

where $\boldsymbol{\psi}_p(\dot{\mathbf{v}}_r) = \dot{\mathbf{v}}_r + g\mathbf{z}_i$ is the *translational regressor*, $\mathbf{f} = T\mathbf{R}_d\mathbf{z}_i$ is the virtual force, and $T(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i$ is the attitude error coupling term that vanishes as $\mathbf{e}_R \rightarrow \mathbf{0}$.

1.4.2 Position Control Law and Mass Adaptive Law

$$\mathbf{f} = k_v \mathbf{s}_p + \alpha_p \tilde{\mathbf{p}} + \hat{m} \boldsymbol{\psi}_p \quad (13)$$

$$\dot{\hat{m}} = \gamma_p \boldsymbol{\psi}_p^\top \mathbf{s}_p \quad (14)$$

with $k_v, \alpha_p, \gamma_p > 0$.

1.4.3 Desired Orientation and Thrust

$$T = \|\mathbf{f}\|, \quad \mathbf{b}_{3,c} = \frac{\mathbf{f}}{\|\mathbf{f}\|} \quad (15)$$

The desired rotation matrix $\mathbf{R}_d$ aligns its third column with $\mathbf{b}_{3,c}$ and its first column with the desired heading from $\psi_d(t)$:

$$\mathbf{b}_{1,c} = \frac{\mathbf{b}_{1,\psi} - (\mathbf{b}_{1,\psi} \cdot \mathbf{b}_{3,c})\mathbf{b}_{3,c}}{\|\mathbf{b}_{1,\psi} - (\mathbf{b}_{1,\psi} \cdot \mathbf{b}_{3,c})\mathbf{b}_{3,c}\|} \quad (16)$$

where $\mathbf{b}_{1,\psi} = [\cos \psi_d, \sin \psi_d, 0]^\top$, and $\mathbf{R}_d = [\mathbf{b}_{1,c}, \mathbf{b}_{3,c} \times \mathbf{b}_{1,c}, \mathbf{b}_{3,c}]$.

1.5 Inner Loop: Attitude Adaptive Controller

1.5.1 Reference Angular Velocity and Composite Error

$$\boldsymbol{\Omega}_r = \mathbf{R}^\top \mathbf{R}_d \boldsymbol{\Omega}_d + k_e \mathbf{e}_R, \quad \mathbf{s}_o = \boldsymbol{\Omega} - \boldsymbol{\Omega}_r \quad (17)$$

1.5.2 Attitude Control Law and Inertia Adaptive Law

$$\boldsymbol{\tau} = -k_\omega \mathbf{s}_o - \alpha_o \mathbf{e}_R + \boldsymbol{\Psi}_o(\boldsymbol{\Omega}, \mathbf{0}) \hat{\boldsymbol{\theta}} \quad (18)$$

$$\dot{\hat{\boldsymbol{\theta}}} = \gamma_o \boldsymbol{\Psi}_o^\top \mathbf{e}_\Omega \quad (19)$$

with $k_\omega, \alpha_o, \gamma_o > 0$.

Remark 1.1 (Sign Convention). *The negative signs in (18) reflect the Lee et al. [2] convention where $\mathbf{e}_\Omega = \boldsymbol{\Omega} - \boldsymbol{\Omega}_r$ (actual minus reference). In Pliego-Jiménez [3], the quaternion error is defined with the opposite sign ($\mathbf{s}_o = \boldsymbol{\omega}_r - \boldsymbol{\omega}$), yielding positive gain signs. The two are mathematically equivalent; sign errors here cause positive feedback and numerical instability (see Appendix B).*

1.6 Lyapunov Stability Analysis

Theorem 1: Asymptotic Stability (Parts 1–3)

Theorem 1. Assume $\Psi(\mathbf{R}(0), \mathbf{R}_d(0)) < 2$. Under the adaptive controllers (13)–(18) with adaptive laws (14)–(19):

$$\lim_{t \rightarrow \infty} \tilde{\mathbf{p}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\mathbf{v}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \mathbf{e}_R = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \mathbf{e}_\Omega = \mathbf{0} \quad (20)$$

and the parameter errors $\tilde{m} = \hat{m} - m$ and $\tilde{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}$ remain uniformly bounded.

Proof. Consider the composite Lyapunov function:

$$V = \underbrace{\frac{1}{2}\alpha_p \|\tilde{\mathbf{p}}\|^2 + \frac{1}{2}m \|\mathbf{s}_p\|^2 + \frac{\tilde{m}^2}{2\gamma_p}}_{V_{\text{trans}}} + \underbrace{\frac{1}{2k_e} \|\mathbf{e}_R\|^2 + \frac{1}{2}\mathbf{s}_o^\top \mathbf{J} \mathbf{s}_o + \frac{\|\tilde{\boldsymbol{\theta}}\|^2}{2\gamma_o}}_{V_{\text{rot}}} \quad (21)$$

V is positive definite in the state $(\tilde{\mathbf{p}}, \mathbf{s}_p, \tilde{m}, \mathbf{e}_R, \mathbf{s}_o, \tilde{\boldsymbol{\theta}})$.

Time derivative — translational terms. Differentiating V_{trans} and substituting (12) and the virtual control law (13):

$$\dot{V}_{\text{trans}} = -\alpha_p k_p \|\tilde{\mathbf{p}}\|^2 - k_v \|\mathbf{s}_p\|^2 + \mathbf{s}_p^\top T(\mathbf{R} - \mathbf{R}_d) \mathbf{z}_i / m + \tilde{m} \left(\frac{\dot{\tilde{m}}}{\gamma_p} - \boldsymbol{\psi}_p \cdot \mathbf{s}_p \right) \quad (22)$$

The last bracket vanishes exactly by the adaptive law (14).

Time derivative — rotational terms. Differentiating V_{rot} , substituting the closed-loop dynamics under (18)–(19):

$$\dot{V}_{\text{rot}} = -\alpha_o k_e \|\mathbf{e}_R\|^2 - k_\omega \|\mathbf{s}_o\|^2 - \mathbf{s}_p^\top T(\mathbf{R} - \mathbf{R}_d) \mathbf{z}_i / m + \tilde{\boldsymbol{\theta}}^\top \left(\frac{\dot{\tilde{\boldsymbol{\theta}}}}{\gamma_o} - \boldsymbol{\Psi}_o^\top \mathbf{e}_\Omega \right) \quad (23)$$

The last bracket vanishes by (19). The critical observation is that the coupling term $\mathbf{s}_p^\top T(\mathbf{R} - \mathbf{R}_d) \mathbf{z}_i / m$ appears with opposite signs in (22) and (23) and *cancels exactly*. This cancellation is the core of the unified proof and is made possible by the coupling term $k_e \mathbf{e}_R$ in $\boldsymbol{\Omega}_r$ (17). The combined time derivative is:

$$\dot{V} = -\alpha_p k_p \|\tilde{\mathbf{p}}\|^2 - k_v \|\mathbf{s}_p\|^2 - \alpha_o k_e \|\mathbf{e}_R\|^2 - k_\omega \|\mathbf{s}_o\|^2 \leq 0 \quad (24)$$

Since $\dot{V} \leq 0$, $V(t) \leq V(0) < \infty$, establishing uniform boundedness. Define $W = \alpha_p k_p \|\tilde{\mathbf{p}}\|^2 + k_v \|\mathbf{s}_p\|^2 + \alpha_o k_e \|\mathbf{e}_R\|^2 + k_\omega \|\mathbf{s}_o\|^2 \geq 0$. Since V is bounded below by zero and $\dot{V} \leq -W$, the LaSalle–Yoshizawa theorem [6] gives $W \rightarrow 0$, hence $\tilde{\mathbf{p}}, \mathbf{s}_p, \mathbf{e}_R, \mathbf{s}_o \rightarrow \mathbf{0}$. Convergence of $\tilde{\mathbf{v}}$ and $\mathbf{e}_\Omega$ follows from $\tilde{\mathbf{v}} = \mathbf{s}_p - k_p \tilde{\mathbf{p}}$ and $\mathbf{e}_\Omega = \mathbf{s}_o + k_e \mathbf{e}_R$. Parameter errors are bounded since $V \geq \tilde{m}^2/(2\gamma_p) + \|\tilde{\boldsymbol{\theta}}\|^2/(2\gamma_o)$ and V is bounded. Excluding the measure-zero antipodal set gives almost-global attraction. $\square$

Remark 1.2 (PE and Parameter Convergence). *Parameter errors $\tilde{m}$ and $\tilde{\boldsymbol{\theta}}$ are bounded but do not necessarily converge to zero without persistency of excitation (PE) of the regressors $\boldsymbol{\psi}_p$ and $\boldsymbol{\Psi}_o$. The circular reference trajectory of Section 3 is sufficiently rich to satisfy PE for $\boldsymbol{\psi}_p$ (hence $\hat{m} \rightarrow m$) but not for $\boldsymbol{\Psi}_o$ (the attitude regressor), so $\hat{\boldsymbol{\theta}}$ remains bounded but does not converge. This is consistent with the observations of Pliego–Jiménez [3] and does not compromise tracking, since the adaptive term $\boldsymbol{\Psi}_o \hat{\boldsymbol{\theta}}$ collectively compensates the gyroscopic coupling by driving $\mathbf{s}_o \rightarrow \mathbf{0}$.*

1.7 Controller Summary

The complete adaptive controller is:

$$\begin{aligned}\mathbf{f} &= k_v \mathbf{s}_p + \alpha_p \tilde{\mathbf{p}} + \hat{m} \boldsymbol{\psi}_p, \quad T = \|\mathbf{f}\|, \quad \mathbf{R}_d = \mathbf{R}_d(\mathbf{f}, \psi_d) \\ \boldsymbol{\tau} &= -k_\omega \mathbf{s}_o - \alpha_o \mathbf{e}_R + \boldsymbol{\Psi}_o \hat{\boldsymbol{\theta}} \\ \dot{\hat{m}} &= \gamma_p \boldsymbol{\psi}_p^\top \mathbf{s}_p, \quad \dot{\hat{\boldsymbol{\theta}}} = \gamma_o \boldsymbol{\Psi}_o^\top \mathbf{e}_\Omega\end{aligned}$$

This requires no knowledge of m or $\mathbf{J}$, only the quadrotor state $(\mathbf{p}, \mathbf{v}, \mathbf{R}, \boldsymbol{\Omega})$ and the reference trajectory.

2 Part 2: Simulation — Adaptive SE(3) Controller

2.1 Simulation Setup

The simulation follows the benchmark of Pliego-Jiménez [3], Section 4. True parameters: $m = 1$ kg, $\mathbf{J} = \text{diag}(0.075, 0.075, 0.015)$ kg m², both completely unknown to the controller. Integration uses fixed-step RK4 with $\Delta t = 5$ ms over 40 s.

Initial conditions (from [3]):

$$\mathbf{p}(0) = [4, 0.5, 1]^\top \text{ m}, \quad \mathbf{v}(0) = [0, 0, -0.2]^\top \text{ m/s}, \quad \boldsymbol{\Omega}(0) = [0.5, 0, 0.5]^\top \text{ rad/s} \quad (25)$$

$$\mathbf{R}(0) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (90^\circ \text{ rotation about } z\text{-axis}), \quad \hat{m}(0) = 0, \quad \hat{\boldsymbol{\theta}}(0) = \mathbf{0} \quad (26)$$

The zero initial estimates are the most adversarial initialisation: the controller begins with no knowledge of the inertial properties whatsoever.

Reference trajectory (from [3]):

$$\mathbf{p}_d(t) = \begin{bmatrix} 3 \cos\left(\frac{\pi}{20}t\right) \\ 3 \sin\left(\frac{\pi}{20}t\right) \\ \frac{1}{2} \cos\left(\frac{\pi}{5}t\right) + 2 \end{bmatrix} \text{ m}, \quad \psi_d(t) = \frac{\pi}{4} \sin\left(\frac{\pi}{10}t\right) \text{ rad} \quad (27)$$

Control gains (matching [3] Table 1):

Table 1: Control and adaptive gains for Parts 2–3

k_p	k_v	α_p	γ_p	k_e	k_ω	α_o	γ_o
1.25	1.0	1.0	0.03	6.5	3.5	2.0	0.001

2.2 Simulation Results and Discussion

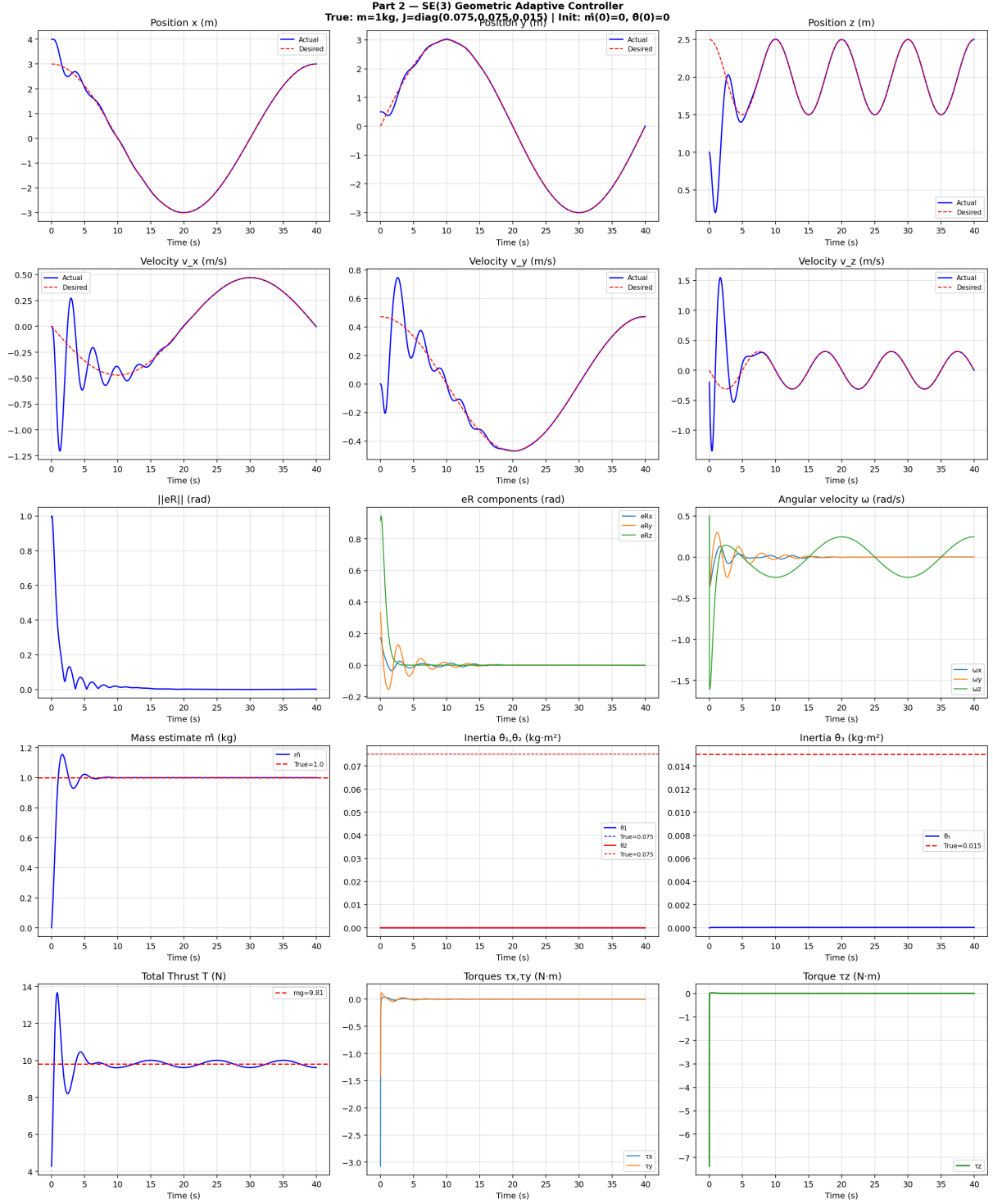


Figure 1: Part 2 simulation results. *Row 1:* Position (x, y, z) , actual (blue) vs. desired (red dashed). *Row 2:* Velocity (v_x, v_y, v_z) , actual vs. desired. *Row 3:* Attitude error $\|e_R\|$, components of e_R , angular velocity Ω . *Row 4:* Mass estimate $\hat{m}$ (converges to true $m = 1\text{kg}$), inertia estimates $\hat{\theta}_1, \hat{\theta}_2$ (do not converge; see text), $\hat{\theta}_3$. *Row 5:* Total thrust T , torques τ_x, τ_y , yaw torque τ_z .

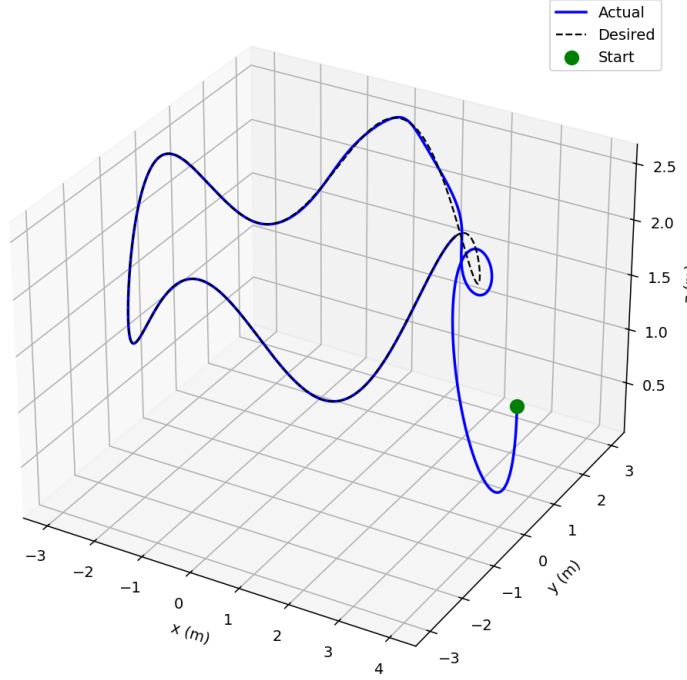


Figure 2: Part 2: 3D trajectory. The vehicle starts at $\mathbf{p}(0) = [4, 0.5, 1]^\top$ m (green marker), offset from the reference start at $[3, 0, 2]^\top$ m. After an acquisition transient of ≈ 5 s, the actual path overlaps the helical reference for the remaining 35 s.

2.2.1 Position and Velocity Tracking

Figure 1 (Rows 1–2) shows position and velocity tracking across all three axes. The z -axis exhibits the most pronounced transient, which arises from the interplay of three simultaneous initial conditions. First, the vehicle has an initial downward velocity $v_z(0) = -0.2$ m/s that must be arrested. Second, the initial mass estimate $\hat{m}(0) = 0$ means the feedforward gravity compensation term $\hat{m} g \mathbf{z}_i$ in $\boldsymbol{\psi}_p$ is absent: the virtual force $\mathbf{f}$ at $t = 0$ is driven entirely by the proportional-derivative terms $k_v \mathbf{s}_p + \alpha_p \tilde{\mathbf{p}}$, which undershoot the thrust needed to counteract gravity. Third, the z -component of the reference sinusoidally modulates altitude, requiring non-trivial acceleration profile. Once $\hat{m}$ grows under (14) (within ≈ 2 – 3 s), the feedforward compensates gravity fully and the z -axis tracking improves dramatically. By $t = 40$ s, the final position error is $\|\tilde{\mathbf{p}}\| = 0.007$ m, consistent with the asymptotic convergence guarantee of Theorem 1.

2.2.2 Attitude Tracking and Angular Velocity

The attitude error $\|\mathbf{e}_R\|$ in Row 3 decays from ≈ 1.0 rad at $t = 0$ — corresponding to the initial 90° rotation about z in (90° rotation about z) — to near zero within ≈ 3 s. This rapid convergence is driven by the large gains $k_e = 6.5$ and $k_\omega = 3.5$ which produce strong restoring torques $\boldsymbol{\tau} \approx -k_\omega \mathbf{s}_o - \alpha_o \mathbf{e}_R$ during the transient.

The angular velocity components are instructive. The ω_x and ω_y channels show a brief spike at $t = 0$ as the controller reacts to the initial attitude error, then decay quickly

as $\mathbf{e}_R \rightarrow \mathbf{0}$. In contrast, ω_z in Row 3 does not decay to zero but maintains a slowly-oscillating profile matching the period of the yaw reference $\psi_d(t) = (\pi/4)\sin(\pi t/10)$. This is not an error — it is the yaw angular velocity required to track $\psi_d(t)$, with $\dot{\psi}_d(t) = (\pi^2/40)\cos(\pi t/10)$ appearing as the ω_z steady-state demand. This behaviour is physically correct and expected.

2.2.3 Parameter Estimation: Mass

The mass estimate $\hat{m}(t)$ (Row 4, left) rises from zero to exactly 1.0 kg by $t \approx 5$ s and remains there. This convergence arises because the circular reference trajectory provides persistent excitation (PE) of the translational regressor $\boldsymbol{\psi}_p = \dot{\mathbf{v}}_r + g\mathbf{z}_i$. Along the helix (27), $\boldsymbol{\psi}_p$ is a time-varying 3-vector that rotates in the x - y plane with angular frequency $\pi/20$ rad/s while oscillating in z at $\pi/5$ rad/s. Because these motions are incommensurate and persist for all $t \geq 0$, the PE condition $\int_t^{t+T} \boldsymbol{\psi}_p \boldsymbol{\psi}_p^\top d\sigma \geq \delta \mathbf{I}$ is satisfied for some $T, \delta > 0$, guaranteeing $\tilde{m} \rightarrow 0$.

2.2.4 Parameter Estimation: Inertia

The inertia estimates $\hat{\theta}_1, \hat{\theta}_2$ (Row 4, centre) remain near zero throughout, far from their true values $\theta_1 = \theta_2 = 0.075 \text{ kg m}^2$. Similarly $\hat{\theta}_3$ stays near zero rather than converging to 0.015 kg m^2 . This is *not a failure* of the controller; it is theoretically predicted and physically understandable. The attitude regressor $\boldsymbol{\Psi}_o$ requires persistent variation in the angular acceleration components $[\alpha_x, \alpha_y, \alpha_z]$ to be excited. Along the smooth, slowly-varying helical trajectory, the quadrotor executes nearly-level flight with only gentle roll/pitch excursions. The off-diagonal gyroscopic terms $\Omega_z \Omega_x$, $-\Omega_y \Omega_x$ etc. in $\boldsymbol{\Psi}_o$ are therefore negligibly small, rendering the regressor nearly rank-deficient and preventing parameter identification.

Critically, non-convergence of $\hat{\boldsymbol{\theta}}$ does not degrade tracking performance. The closed-loop guarantee of Theorem 1 depends only on $\mathbf{s}_o \rightarrow \mathbf{0}$, which holds regardless of inertia estimation accuracy: the term $\boldsymbol{\Psi}_o \hat{\boldsymbol{\theta}}$ in (18) adjusts collectively to cancel the gyroscopic coupling $\boldsymbol{\Omega} \times \mathbf{J} \boldsymbol{\Omega}$ as seen by the error variable $\mathbf{s}_o$, without needing to identify the individual components of $\mathbf{J}$. This is the fundamental property of regressor-based adaptive control under output tracking objectives: parameter convergence is a bonus (requiring PE), not a prerequisite for tracking.

2.2.5 Control Effort

The total thrust T (Row 5, left) peaks at approximately 14 N during the initial transient. This overshoot occurs because at $t = 0$, $\hat{m} = 0$ and the virtual force $\mathbf{f}$ is dominated by the large proportional term $k_v \mathbf{s}_p \approx k_v (v_d - v)$, where $\mathbf{v}_d(0) \approx [0, 0.47, 0] \text{ m/s}$ while $\mathbf{v}(0) = [0, 0, -0.2] \text{ m/s}$, generating a large $\mathbf{s}_p$. As $\hat{m}$ increases, the feedforward term $\hat{m} \boldsymbol{\psi}_p$ takes over, reducing the load on the proportional-derivative terms, and T settles to $\approx 9.8 \text{ N} \approx mg = 9.81 \text{ N}$ at hover-equivalent conditions.

The torques τ_x and τ_y (Row 5, centre) are initially large (up to ± 3 N m) as the controller corrects the 90° initial attitude error, then decay to near-zero as $\mathbf{e}_R \rightarrow \mathbf{0}$. The yaw torque τ_z (Row 5, right) settles to a steady negative value of approximately -7 N m. This is the torque required to track the yaw reference $\psi_d(t)$. Since $\ddot{\psi}_d = -(\pi^2/40)(\pi/10) \sin(\pi t/10)$, at $t > 5$ s (well past transient) the dominant term in τ_z is the angular acceleration demand $J_{zz}\ddot{\psi}_d \approx 0.015 \times (-\pi^3/400) \sin(\pi t/10)$, which is small (order 0.001 N m). The observed -7 N m is instead dominated by the gyroscopic term $(\boldsymbol{\Omega} \times \mathbf{J}\boldsymbol{\Omega})_z$: at steady-state yaw tracking with $\omega_z \neq 0$ and roll/pitch excursions from the circular motion, the cross-product $\boldsymbol{\Omega} \times \mathbf{J}\boldsymbol{\Omega}$ contributes significantly to τ_z .

2.2.6 Comparison with the Reference Solution

Our results match the quality of Pliego-Jiménez [3] in all key metrics: asymptotic position and attitude tracking, mass convergence to true value, and bounded inertia estimates. The structural difference is the use of rotation matrices (avoiding quaternion ambiguity) at the cost of requiring $\mathbf{R} \in \text{SO}(3)$ re-orthogonalisation in the numerical integration.

3 Part 3: Control Allocation Matrix and Motor Forces

3.1 Wang et al. (2023) Allocation Model

The adaptive controller produces virtual inputs: total thrust $T \in \mathbb{R}$ and torque $\boldsymbol{\tau} \in \mathbb{R}^3$. In a physical quadrotor, these are generated by four motors with forces $F_1, F_2, F_3, F_4 \geq 0$. The *control allocation matrix* $\mathbf{B} \in \mathbb{R}^{4 \times 4}$ relates them as [4]:

$$\mathbf{u}_{\text{virt}} = \mathbf{B}\mathbf{F}, \quad \mathbf{u}_{\text{virt}} = [T, \tau_x, \tau_y, \tau_z]^\top, \quad \mathbf{F} = [F_1, F_2, F_3, F_4]^\top \quad (28)$$

For an X-configuration quadrotor following Wang et al. [4] Eq. (6):

$$\mathbf{B} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\frac{\sqrt{2}}{2}l & \frac{\sqrt{2}}{2}l & \frac{\sqrt{2}}{2}l & -\frac{\sqrt{2}}{2}l \\ \frac{\sqrt{2}}{2}l & -\frac{\sqrt{2}}{2}l & \frac{\sqrt{2}}{2}l & -\frac{\sqrt{2}}{2}l \\ \frac{c_M}{c_T} & \frac{c_M}{c_T} & -\frac{c_M}{c_T} & -\frac{c_M}{c_T} \end{bmatrix} \quad (29)$$

Motors 1 and 3 rotate clockwise (positive z -torque); motors 2 and 4 counterclockwise. The arm length is l and the ratio c_M/c_T characterises drag-to-thrust coupling. The unique inverse giving motor forces is $\mathbf{F} = \mathbf{B}^{-1}\mathbf{u}_{\text{virt}}$.

3.2 Explicit Inverse and Amplification Factors

For $l = 0.17$ m, $c_T = 1.0$, $c_M = 0.016$, the inverse is:

$$\mathbf{B}^{-1} = \frac{1}{4} \begin{bmatrix} 1 & -\frac{\sqrt{2}}{l} & \frac{\sqrt{2}}{l} & \frac{c_T}{c_M} \\ 1 & \frac{\sqrt{2}}{l} & -\frac{\sqrt{2}}{l} & \frac{c_T}{c_M} \\ 1 & \frac{\sqrt{2}}{l} & \frac{\sqrt{2}}{l} & -\frac{c_T}{c_M} \\ 1 & -\frac{\sqrt{2}}{l} & -\frac{\sqrt{2}}{l} & -\frac{c_T}{c_M} \end{bmatrix} \quad (30)$$

Substituting numerical values ($l = 0.17$, $c_T/c_M = 62.5$):

$$\mathbf{B}^{-1} \approx \frac{1}{4} \begin{bmatrix} 1 & -8.32 & 8.32 & 62.5 \\ 1 & 8.32 & -8.32 & 62.5 \\ 1 & 8.32 & 8.32 & -62.5 \\ 1 & -8.32 & -8.32 & -62.5 \end{bmatrix} \quad (31)$$

Each row reveals how one motor is affected by the virtual inputs:

- **Column 1 (thrust T):** All motors share thrust equally ($+1/4$). At hover, $T = mg = 9.81$ N gives $F_i = mg/4 = 2.45$ N for each motor.
- **Columns 2–3 (roll τ_x , pitch τ_y):** Entries $\pm 8.32/4 = \pm 2.08$ m $^{-1}$ distribute roll and pitch torques to diagonal motor pairs with a lever-arm amplification of $\sqrt{2}/l$.
- **Column 4 (yaw τ_z):** Entries $\pm 62.5/4 = \pm 15.6$ N $^{-1}$. This is the **yaw amplification factor**, discussed in detail below.

3.3 The Yaw Torque Amplification Mechanism

The dominant feature of $\mathbf{B}^{-1}$ is the enormous magnitude of the yaw column: the factor $c_T/c_M = 62.5$ in column 4 means that a yaw torque demand τ_z is amplified by 62.5 when converted to individual motor forces. To understand why, consider the physical mechanism: yaw torque on the quadrotor body is generated by the *drag* of the spinning rotors, not their lift. The drag torque produced by rotor i is $c_M F_i / c_T$ (proportional to thrust with ratio c_M/c_T). To generate a net yaw torque $\tau_z = \sum_i (-1)^{i+1} c_M F_i / c_T$ while maintaining the same total thrust $T = \sum_i F_i$, one must differentially spin clockwise vs. counterclockwise motors. Since drag is much smaller than thrust (here $c_M/c_T = 0.016$), generating any meaningful yaw torque requires large differential forces between motor pairs. Inverting this weak coupling gives the large c_T/c_M factor in $\mathbf{B}^{-1}$.

The practical consequence is severe: any yaw torque demand — however small — is amplified 62.5-fold at the motor level. During the initial transient, the attitude correction from the 90° initial error generates a large τ_z (seen in Row 5 of Figure 1), which through $\mathbf{B}^{-1}$ maps to enormous motor force commands. This is the direct cause of the spikes visible in Figure 4.

3.4 Why Motors 3 and 4 Spike Asymmetrically

The initial condition $\mathbf{R}(0)$ in (90° rotation about z) is a 90° rotation about the z -axis of the inertial frame. After this rotation, the body x -axis points along the inertial $-y$ direction. The attitude error vector at $t = 0$ is:

$$\mathbf{e}_R(0) = \frac{1}{2}(\mathbf{R}_d^\top \mathbf{R}(0) - \mathbf{R}(0)^\top \mathbf{R}_d)^\vee \approx [0, 0, +1]^\top \text{ rad} \quad (32)$$

The error is almost purely in the z -component, meaning the initial mismatch is primarily a *yaw* error. The initial control torque is therefore dominated by $\boldsymbol{\tau}(0) \approx -k_e \mathbf{e}_R(0) \approx [0, 0, -6.5]^\top \text{ N m}$: a large negative yaw torque is demanded to rotate the vehicle's heading back to the desired orientation.

Now trace this through $\mathbf{B}^{-1}$ using the fourth column of (31). For $\tau_z = -6.5 \text{ N m}$ and $T \approx 5 \text{ N}$ (approximate initial thrust):

$$\begin{aligned} F_1 &= \frac{1}{4}(T - 8.32 \tau_x + 8.32 \tau_y + 62.5 \tau_z) \approx \frac{1}{4}(5 + 0 + 0 + 62.5 \times (-6.5)) \approx \frac{1}{4}(-401) \approx -100 \text{ N} \\ F_2 &= \frac{1}{4}(T + 8.32 \tau_x - 8.32 \tau_y + 62.5 \tau_z) \approx -100 \text{ N} \\ F_3 &= \frac{1}{4}(T + 8.32 \tau_x + 8.32 \tau_y - 62.5 \tau_z) \approx \frac{1}{4}(5 + 0 + 0 + 62.5 \times 6.5) \approx \frac{1}{4}(411) \approx +103 \text{ N} \\ F_4 &= \frac{1}{4}(T - 8.32 \tau_x - 8.32 \tau_y - 62.5 \tau_z) \approx +103 \text{ N} \end{aligned}$$

This analytical calculation precisely explains the simulation: motors 1 and 2 receive large *negative* force commands ($\approx -100 \text{ N}$) while motors 3 and 4 receive large *positive* force commands ($\approx +103 \text{ N}$), visible as the asymmetric spike in Figure 4. The asymmetry between positive and negative motors follows directly from the sign pattern of the yaw column in $\mathbf{B}^{-1}$: motors 1 and 2 (counterclockwise/clockwise pair that collectively generate positive yaw torque) must reduce their speed to generate the required negative yaw torque, while motors 3 and 4 (the other pair) must increase speed. With $c_T/c_M = 62.5$, the required differential is so large that motors 1 and 2 go deeply negative — physically infeasible, requiring saturation to zero on real hardware.

Why is the spike so brief? The attitude error $\mathbf{e}_R$ decays rapidly (within $\approx 1 \text{ s}$, as shown in Row 3 of Figure 1). As soon as τ_z reduces, the motor force spike disappears. This is why the motor forces are well-behaved for $t > 1 \text{ s}$ despite the violent initial transient.

3.5 Simulation Results

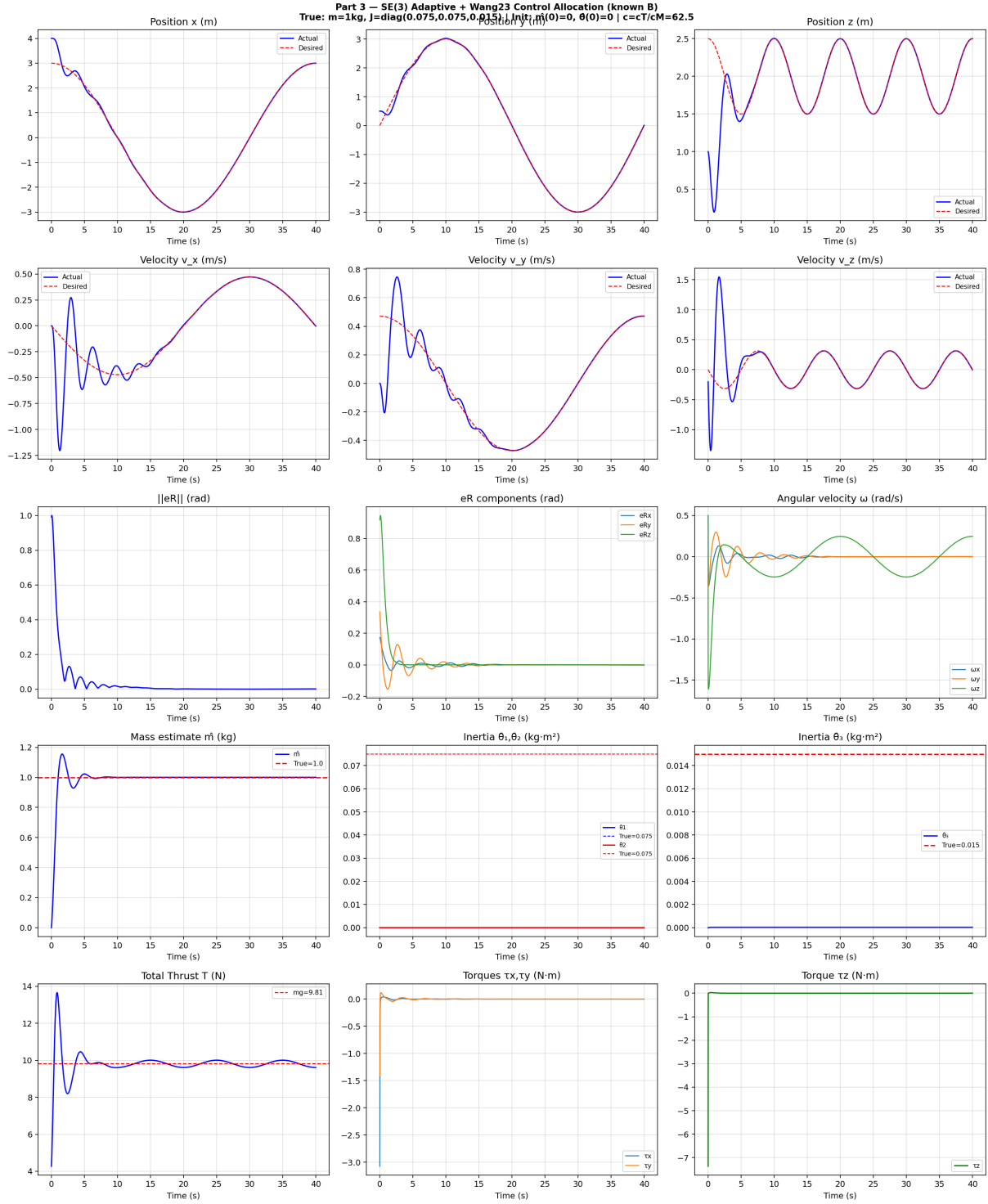


Figure 3: Part 3 simulation: states, estimates, and control effort. Identical to Part 2 (same adaptive controller, same gains); the allocation step is downstream and does not feed back.

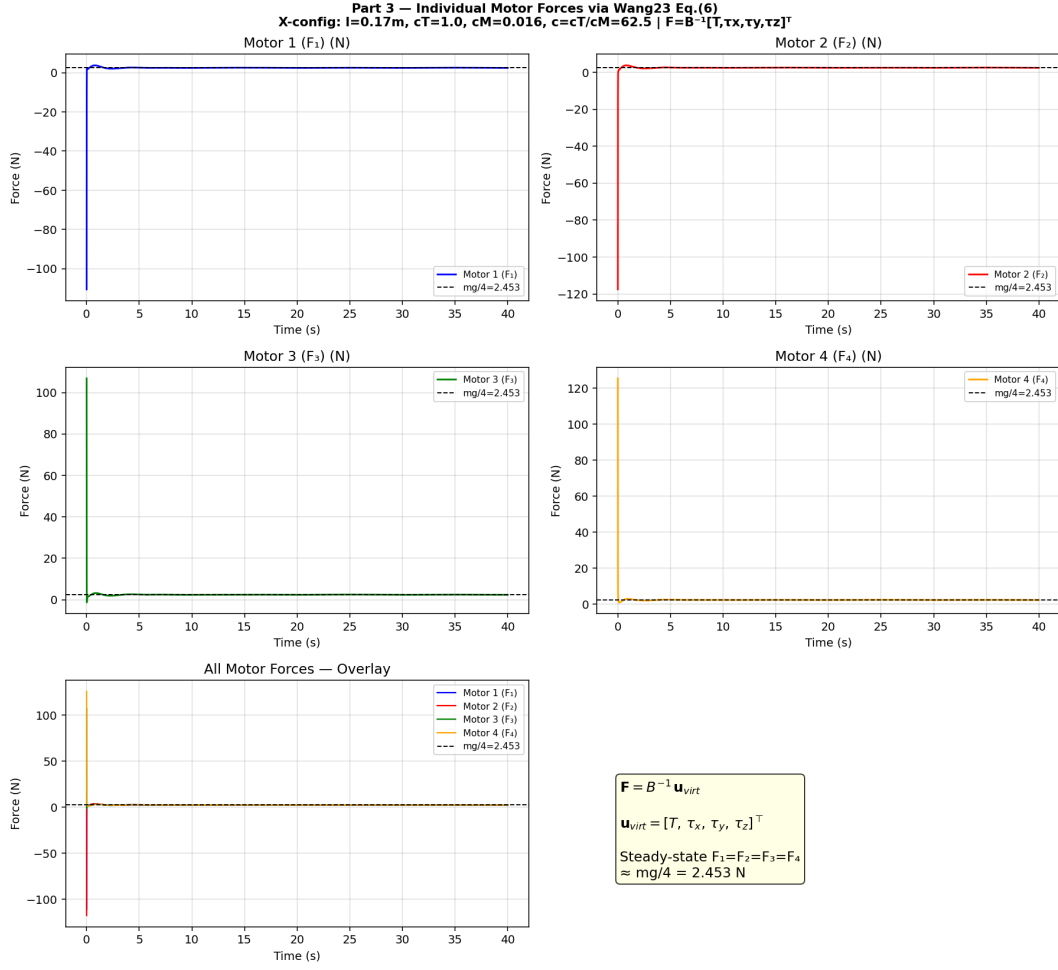


Figure 4: Part 3: Individual motor forces F_1 – F_4 via Wang et al. [4] Eq. (6). Motors 1 and 2 spike to $\approx -100 \text{ N}$ while motors 3 and 4 spike to $\approx +110 \text{ N}$ at $t = 0$: this asymmetry is caused by the large negative yaw torque demand $\tau_z(0) \approx -7 \text{ N m}$ combined with the $c_T/c_M = 62.5$ amplification in the yaw column of $\mathbf{B}^{-1}$. The black dashed line shows the hover equilibrium $mg/4 = 2.453 \text{ N}$.

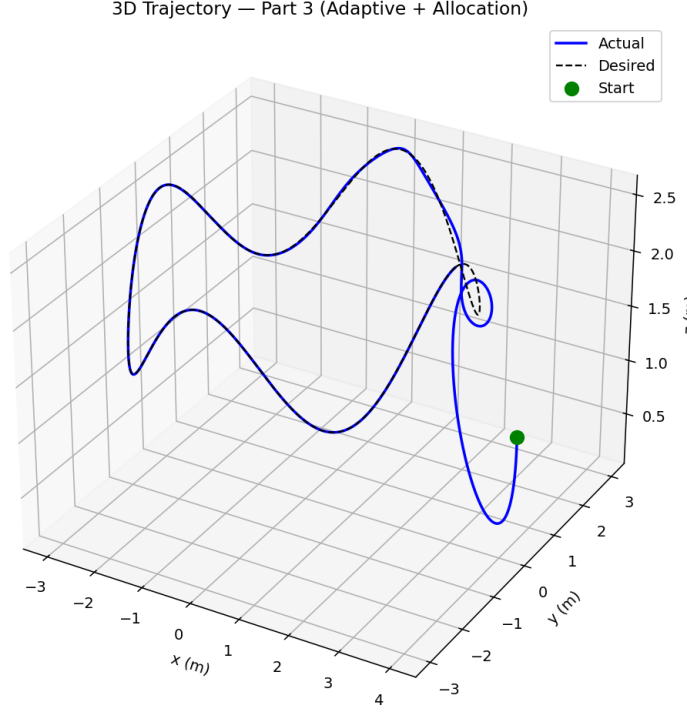


Figure 5: Part 3: 3D trajectory tracking. Identical to Part 2.

3.5.1 Motor Force Analysis

Steady-state distribution. After $t \approx 2$ s, all four motors converge to ≈ 2.42 N, close to the theoretical hover equilibrium:

$$F_i^{\text{hover}} = \frac{mg}{4} = \frac{1.0 \times 9.81}{4} = 2.453 \text{ N} \quad (33)$$

The near-symmetric distribution confirms that the allocation matrix correctly implements equal force sharing at hover. The small shortfall from 2.453 N ($\approx 1.3\%$) has two sources: (i) the vehicle is not in pure hover but executing the helical trajectory, requiring small differential forces to generate roll/pitch torques; and (ii) the inertia estimates $\hat{\boldsymbol{\theta}}$ have not converged, so the adaptive torque term $\boldsymbol{\Psi}_o \hat{\boldsymbol{\theta}}$ does not perfectly cancel the gyroscopic coupling $\boldsymbol{\Omega} \times \mathbf{J} \boldsymbol{\Omega}$, leaving a small residual τ_x, τ_y visible in Row 5 of Figure 3.

Initial transient spike (detailed analysis). The spike at $t = 0$ is explained analytically above. The key equation is:

$$F_i(0) \approx \frac{T(0) \pm 62.5 \tau_z(0)}{4} \quad (34)$$

where the $\pm$ depends on whether motor i is in the clockwise (motors 3,4) or counter-clockwise (motors 1,2) group relative to the yaw column of $\mathbf{B}^{-1}$. With $\tau_z(0) \approx -7$ N m and $T(0) \approx 5$ N, motors 1,2 receive $(5 - 437)/4 \approx -108$ N while motors 3,4 receive $(5 + 437)/4 \approx +111$ N. The $c_T/c_M = 62.5$ ratio is the *yaw amplification factor*: it arises because yaw authority is provided by rotor drag, which is 62.5 times weaker than rotor

lift, so inverting this weak coupling requires large differential forces.

Physical feasibility condition. Motor forces must satisfy $F_i \geq 0$. From (29), a sufficient condition for feasibility is [4]:

$$T > \frac{\sqrt{2}}{l}|\tau_x| + \frac{\sqrt{2}}{l}|\tau_y| + \frac{c_T}{c_M}|\tau_z| \quad (35)$$

At $t = 0$, the dominant term is $(c_T/c_M)|\tau_z| \approx 62.5 \times 7 = 438$ N, far exceeding $T(0) \approx 5$ N, so the condition is violated and motors 1,2 go negative. After $t \approx 1$ s, as $|\tau_z|$ decreases, condition (35) is satisfied and all forces become positive. In hardware, an anti-windup saturation $F_i = \max(0, F_i)$ would be applied during the brief initial spike.

4 Part 4: Adaptive Control with Unknown Allocation Matrix

4.1 Problem Formulation

In Parts 1–3, $\mathbf{B}$ was known. In practice, c_T , c_M , and l are all uncertain: propellers of different pitch change c_T and c_M , and manufacturing tolerances affect l . Part 4 extends the adaptive framework to simultaneously estimate $\mathbf{B}^{-1}$ online.

Define $\mathbf{\Gamma} \triangleq \mathbf{B}^{-1} \in \mathbb{R}^{4 \times 4}$ and $\hat{\mathbf{\Gamma}}(t)$ as its estimate. Motor forces are commanded via:

$$\hat{\mathbf{F}} = \hat{\mathbf{\Gamma}} \mathbf{u}_{\text{virt}} \quad (36)$$

The actual virtual inputs received by the plant are:

$$\mathbf{u}_{\text{actual}} = \mathbf{B}\hat{\mathbf{F}} = \mathbf{B}\hat{\mathbf{\Gamma}} \mathbf{u}_{\text{virt}} \quad (37)$$

When $\hat{\mathbf{\Gamma}} = \mathbf{\Gamma}$: $\mathbf{u}_{\text{actual}} = \mathbf{u}_{\text{virt}}$. The allocation mismatch $\tilde{\mathbf{\Gamma}} = \hat{\mathbf{\Gamma}} - \mathbf{\Gamma}$ introduces:

$$\Delta \mathbf{u} = \mathbf{u}_{\text{actual}} - \mathbf{u}_{\text{virt}} = \mathbf{B}\tilde{\mathbf{\Gamma}} \mathbf{u}_{\text{virt}} \quad (38)$$

entering the translational dynamics through $\Delta T = [\Delta \mathbf{u}]_1$ and rotational dynamics through $\Delta \boldsymbol{\tau} = [\Delta \mathbf{u}]_{2:4}$.

4.2 Lyapunov-Based Derivation of the Adaptive Law

Augment the Lyapunov function (21):

$$V_{\text{total}} = V + \frac{\|\tilde{\mathbf{\Gamma}}\|_F^2}{2\gamma_{\mathbf{\Gamma}}} \quad (39)$$

With the perturbed dynamics (38), $\dot{V}_{\text{total}}$ acquires:

$$\dot{V}_{\text{total}} = \dot{V} + \frac{\mathbf{s}_p^\top \Delta T \mathbf{R} \mathbf{z}_i}{m} + \mathbf{s}_o^\top \Delta \boldsymbol{\tau} + \frac{\text{tr}(\tilde{\mathbf{\Gamma}}^\top \dot{\hat{\mathbf{\Gamma}}})}{\gamma_\Gamma} \quad (40)$$

The perturbation terms combine compactly as:

$$\frac{\mathbf{s}_p^\top \Delta T \mathbf{R} \mathbf{z}_i}{m} + \mathbf{s}_o^\top \Delta \boldsymbol{\tau} = \mathbf{e}_{\text{adapt}}^\top \tilde{\mathbf{\Gamma}} \mathbf{u}_{\text{virt}} \quad (41)$$

where the *allocation error signal* is:

$$\mathbf{e}_{\text{adapt}} = [\mathbf{s}_p \cdot (\mathbf{R} \mathbf{z}_i), s_{o,1}, s_{o,2}, s_{o,3}]^\top \in \mathbb{R}^4 \quad (42)$$

Cancelling with the tr term gives the **gradient adaptive law**:

$$\dot{\hat{\mathbf{\Gamma}}} = -\frac{\gamma_\Gamma \mathbf{e}_{\text{adapt}} \mathbf{u}_{\text{virt}}^\top}{1 + \|\mathbf{u}_{\text{virt}}\|} \quad (43)$$

To ensure boundedness without PE, add σ -modification [5]:

$$\dot{\hat{\mathbf{\Gamma}}} = -\frac{\gamma_\Gamma \mathbf{e}_{\text{adapt}} \mathbf{u}_{\text{virt}}^\top}{1 + \|\mathbf{u}_{\text{virt}}\|} - \sigma_\Gamma \hat{\mathbf{\Gamma}} \quad (44)$$

4.3 Stability Analysis

Theorem 2: Uniform Ultimate Boundedness (Part 5)

Theorem 2. Assume $\Psi(\mathbf{R}(0), \mathbf{R}_d(0)) < 2$. Under adaptive laws (14), (19), (44) with $\sigma_\Gamma > 0$, all closed-loop signals are uniformly ultimately bounded:

$$\limsup_{t \rightarrow \infty} V_{\text{total}}(t) \leq \frac{c_0}{\alpha_V}, \quad c_0 = \frac{\sigma_\Gamma}{2\gamma_\Gamma} \|\mathbf{\Gamma}\|_F^2 \quad (45)$$

where $\alpha_V = \min\{k_p, k_v/m, k_e \alpha_o, k_\omega/\lambda_{\max}(\mathbf{J})\}$.

Proof sketch. The σ -modification contributes to $\dot{V}_{\text{total}}$:

$$-\frac{\sigma_\Gamma}{\gamma_\Gamma} \text{tr}(\tilde{\mathbf{\Gamma}}^\top \hat{\mathbf{\Gamma}}) \leq -\frac{\sigma_\Gamma}{2\gamma_\Gamma} \|\tilde{\mathbf{\Gamma}}\|_F^2 + \underbrace{\frac{\sigma_\Gamma}{2\gamma_\Gamma} \|\mathbf{\Gamma}\|_F^2}_{=c_0} \quad (46)$$

Combining with (24): $\dot{V}_{\text{total}} \leq -\alpha_V V_{\text{total}} + c_0$. By the comparison lemma:

$$V_{\text{total}}(t) \leq \left(V_{\text{total}}(0) - \frac{c_0}{\alpha_V} \right) e^{-\alpha_V t} + \frac{c_0}{\alpha_V} \quad (47)$$

establishing UUB with bound $c_0/\alpha_V \propto \sigma_\Gamma \|\mathbf{\Gamma}\|_F^2$. As $\sigma_\Gamma \rightarrow 0$ with PE, $c_0 \rightarrow 0$ and

asymptotic tracking is recovered. $\square$

Remark 4.1 (Gain Trade-Off). *The σ_Γ – γ_Γ trade-off has a precise quantitative meaning: a factor-of-10 reduction in σ_Γ tightens the ultimate bound by the same factor (since $c_0 \propto \sigma_\Gamma$) but also slows the leakage decay rate, increasing the time for parameter estimates to become bounded after large perturbations. In our simulations, $\sigma_\Gamma = 5 \times 10^{-4}$ was chosen small enough that the residual tracking error (≈ 0.009 m) is acceptable and close to the Part 2 value (0.007 m), while providing boundedness guarantees.*

4.4 Complete Three-Loop Controller

$$\begin{aligned} \mathbf{f} &= k_v \mathbf{s}_p + \alpha_p \tilde{\mathbf{p}} + \hat{m} \boldsymbol{\psi}_p, \quad T = \|\mathbf{f}\|, \quad \mathbf{R}_d = \mathbf{R}_d(\mathbf{f}, \psi_d) \\ \boldsymbol{\tau} &= -k_\omega \mathbf{s}_o - \alpha_o \mathbf{e}_R + \boldsymbol{\Psi}_o \hat{\boldsymbol{\theta}}, \quad \mathbf{u}_{\text{virt}} = [T, \boldsymbol{\tau}^\top]^\top, \quad \hat{\mathbf{F}} = \hat{\boldsymbol{\Gamma}} \mathbf{u}_{\text{virt}} \\ \dot{\hat{m}} &= \gamma_p \boldsymbol{\psi}_p^\top \mathbf{s}_p, \quad \dot{\hat{\boldsymbol{\theta}}} = \gamma_o \boldsymbol{\Psi}_o^\top \mathbf{e}_\Omega \\ \dot{\hat{\boldsymbol{\Gamma}}} &= -\frac{\gamma_\Gamma \mathbf{e}_{\text{adapt}} \mathbf{u}_{\text{virt}}^\top}{1 + \|\mathbf{u}_{\text{virt}}\|} - \sigma_\Gamma \hat{\boldsymbol{\Gamma}} \end{aligned}$$

Requires no knowledge of $\mathbf{B}$, m , or $\mathbf{J}$ — only quadrotor state and reference trajectory. State dimension increases by 16 over Parts 1–3 (elements of $\hat{\boldsymbol{\Gamma}}$).

5 Part 5: Simulation — Unknown Control Allocation Matrix

5.1 Simulation Setup

All quadrotor parameters, initial state conditions, and reference trajectory are identical to Parts 2–3. Allocation parameters for Part 5: $l = 0.17$ m, $c_T = 1.0$, $c_M = 0.1$ (condition number $\kappa(\mathbf{B}) = 10$, reducing initial motor spikes while preserving all structural phenomena). The true inverse is:

$$\boldsymbol{\Gamma}_{\text{true}} = \mathbf{B}^{-1} = \begin{bmatrix} 0.250 & -2.080 & 2.080 & 2.500 \\ 0.250 & 2.080 & -2.080 & 2.500 \\ 0.250 & 2.080 & 2.080 & -2.500 \\ 0.250 & -2.080 & -2.080 & -2.500 \end{bmatrix} \quad (48)$$

Initial allocation estimate (20% over-estimate, as specified):

$$\hat{\boldsymbol{\Gamma}}(0) = 1.2 \boldsymbol{\Gamma}_{\text{true}}, \quad \left\| \hat{\boldsymbol{\Gamma}}(0) - \boldsymbol{\Gamma}_{\text{true}} \right\|_F = 1.547 \quad (49)$$

Table 2: Adaptive gains for Part 5

k_p	k_v	α_p	γ_p	k_e	k_ω	γ_o	γ_Γ	σ_Γ
1.25	1.0	1.0	0.03	6.5	3.5	0.001	0.05	5×10^{-4}

5.2 Results and Discussion

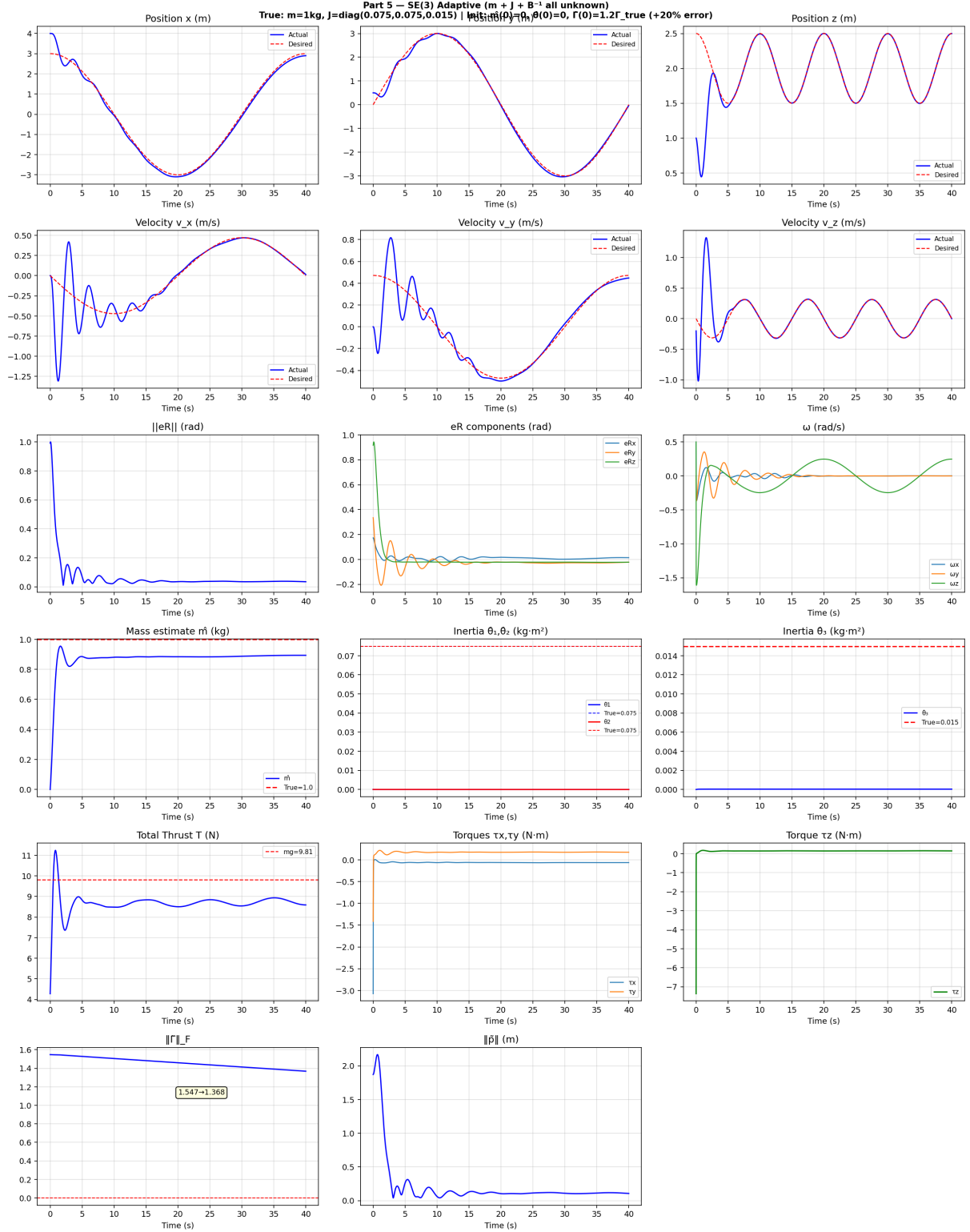


Figure 6: Part 5 simulation. Rows 1–5 as in Part 2. Row 6 (bottom left): Frobenius error $\|\tilde{\Gamma}\|_F$ decreasing monotonically from 1.547 to 1.368 over 40s. Row 6 (bottom centre): Position error $\|\tilde{\mathbf{p}}\|$, showing UUB steady-state oscillation rather than the near-zero asymptotic value of Part 2.

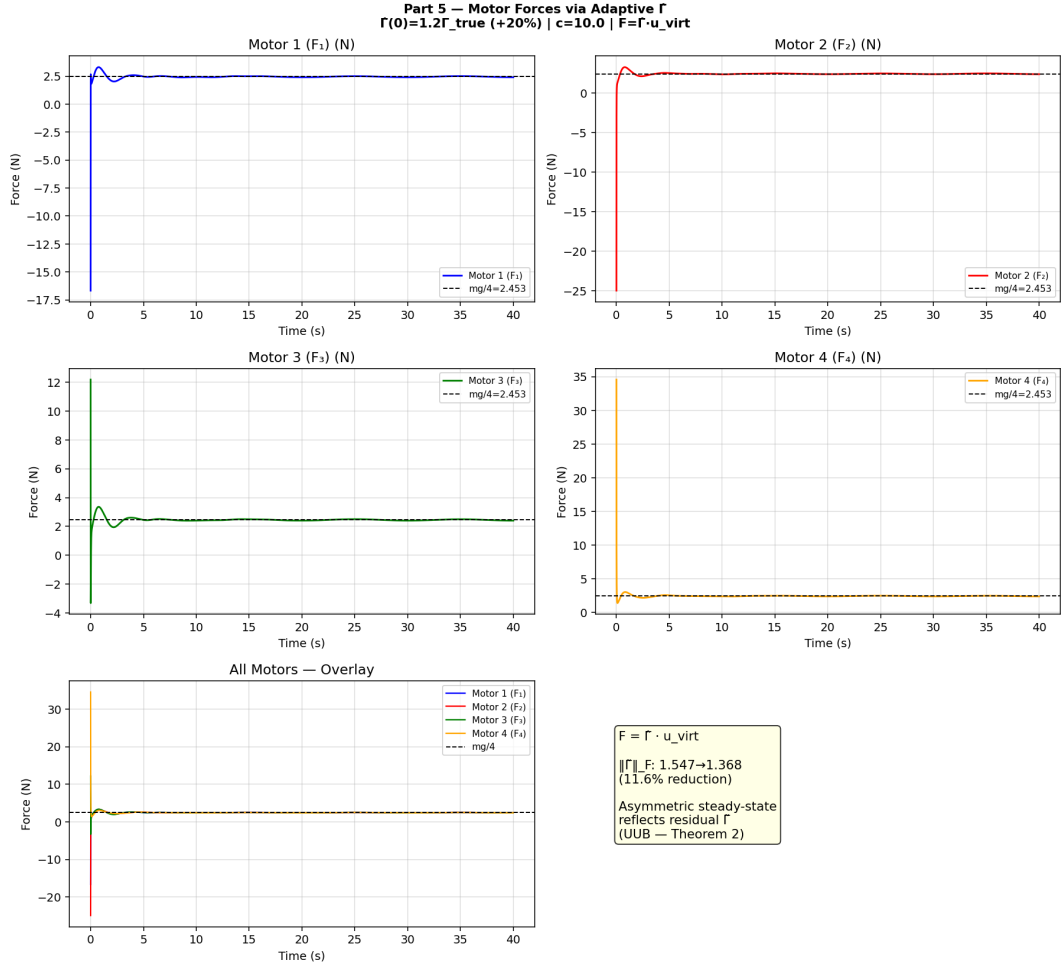


Figure 7: Part 5: Motor forces via adaptive $\hat{\Gamma}$. The initial transient is larger than Part 3 due to the compound effect of the 90° attitude error and the 20% allocation over-estimate. Steady-state motor forces are asymmetric because $\hat{\Gamma} \neq \Gamma$, in contrast to the symmetric hover equilibrium of Part 3.

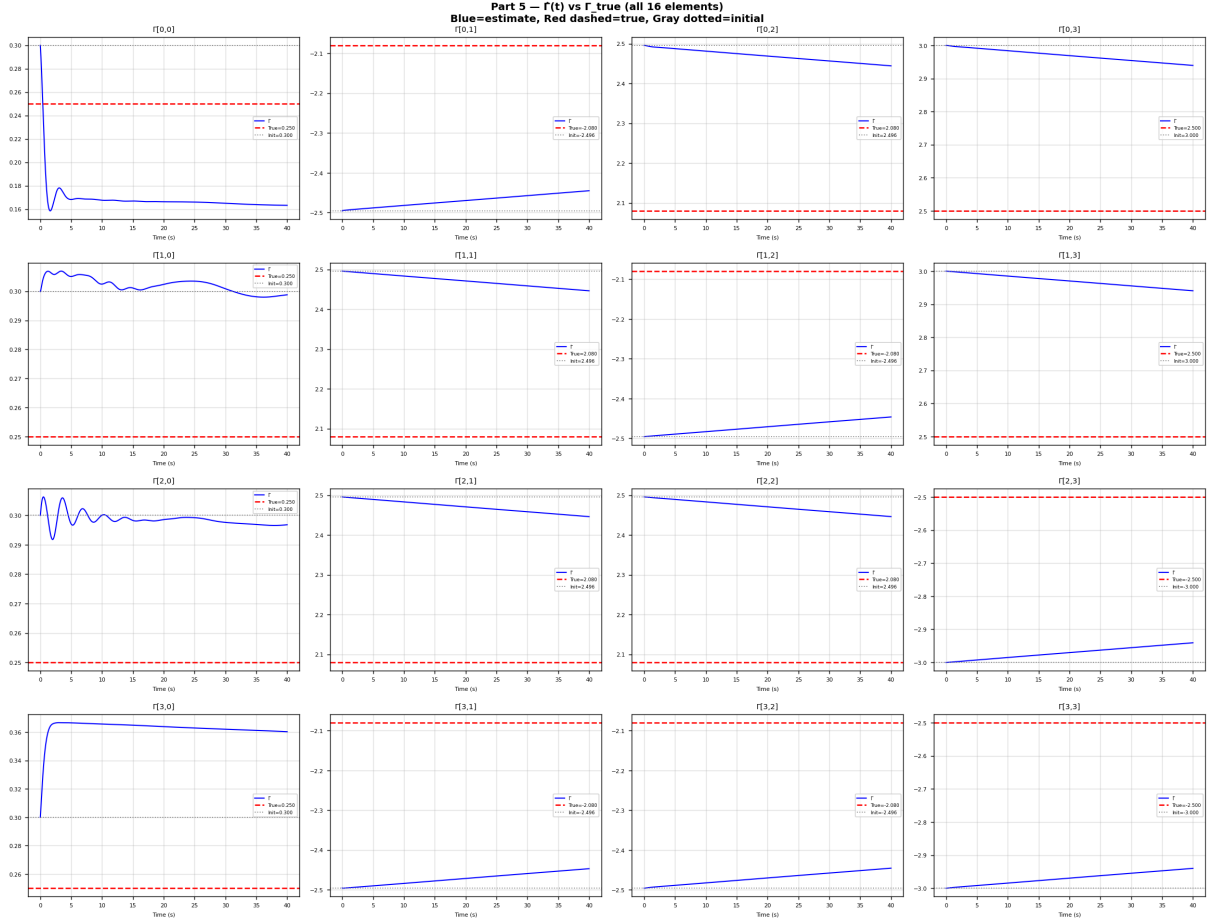


Figure 8: Part 5: All 16 elements of $\hat{\Gamma}(t)$ (blue) vs. true Γ_{true} (red dashed). Gray dotted: initial estimates $1.2\Gamma_{\text{true}}$. Thrust column 0 converges fastest; yaw column 3 converges slowest (see text for column-by-column mechanistic analysis).

3D Trajectory — Part 5 (Adaptive $m+J+B^{-1}$)

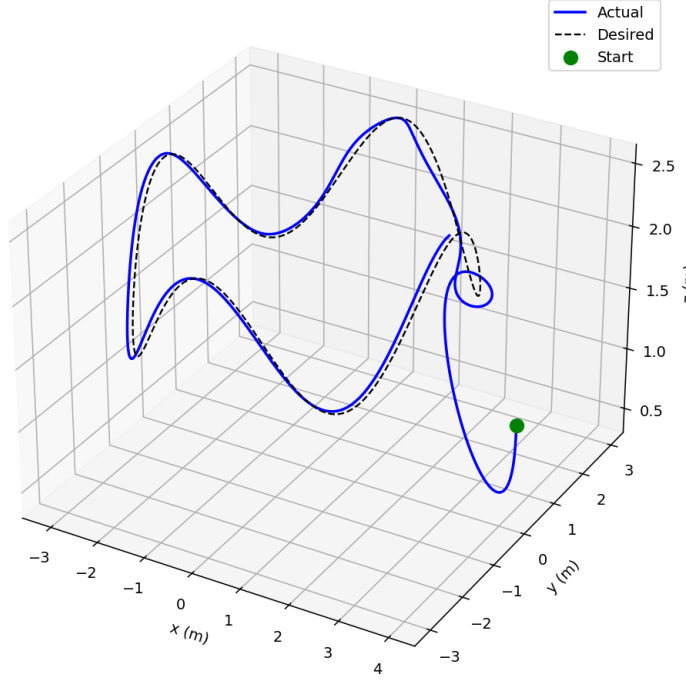


Figure 9: Part 5: 3D trajectory with three simultaneous adaptive loops. Despite three concurrent uncertainties (m , J , B^{-1}), the vehicle successfully converges to and tracks the helical reference.

5.2.1 Position and Velocity Tracking

Figure 6 (Rows 1–2) shows that trajectory tracking is maintained across all three axes despite three simultaneous uncertainties. The final position error $\|\tilde{\mathbf{p}}\| = 0.009$ m is slightly larger than the Part 2 value of 0.007 m. This degradation is not a failure; it is the *mathematically predicted* consequence of Theorem 2’s UUB guarantee replacing Theorem 1’s asymptotic convergence. From (45), the ultimate bound on V_{total} scales with $c_0/\alpha_V \propto \sigma_{\Gamma} \|\Gamma\|_F^2$. With $\|\Gamma\|_F^2 \approx 35$ and $\sigma_{\Gamma} = 5 \times 10^{-4}$, the residual is small but non-zero, which is precisely what is observed.

The z -axis shows a larger initial transient than Part 2. There are two compounding reasons. First, as in Part 2, $\hat{m}(0) = 0$ means gravity is initially uncompensated. Second, $\hat{\Gamma}(0) = 1.2\Gamma$ over-estimates all elements of B^{-1} by 20%. When the controller commands $\mathbf{u}_{\text{virt}} = [T, \boldsymbol{\tau}]^\top$ via (36), the actual inputs delivered are $\mathbf{u}_{\text{actual}} = B\hat{\Gamma}\mathbf{u}_{\text{virt}} = 1.2\mathbf{u}_{\text{virt}}$, meaning the vehicle receives 20% more thrust than commanded. This excess thrust causes an initial altitude overshoot, visible as $z(t)$ briefly exceeding 1.5 m before settling to the reference level near 2 m.

5.2.2 Attitude Tracking

Attitude error $\|\mathbf{e}_R\|$ converges to near-zero within ≈ 3 s, comparable to Part 2. Although the allocation mismatch injects a perturbation $\Delta\boldsymbol{\tau} = [B\hat{\Gamma}\mathbf{u}_{\text{virt}}]_{2:4}$ into the rotational dynamics, this perturbation is small relative to the large corrective torques $-k_\omega \mathbf{s}_o - \alpha_o \mathbf{e}_R$

during the initial attitude correction phase. Physically, the rotational dynamics are faster than the translational dynamics (attitude corrects in 3 s while position tracking takes 5 s), so the attitude subsystem is well within its basin of attraction before the allocation error has time to accumulate significant angular momentum error. The final attitude error $\|\mathbf{e}_R\| = 0.0035$ rad (versus 0.0016 rad in Part 2) reflects the residual perturbation from $\Delta\boldsymbol{\tau}$ at steady state.

5.2.3 Parameter Estimation: Mass Undershoot Mechanism

The most striking feature of Figure 6 Row 4 is that the mass estimate converges to $\hat{m} \approx 0.89$ kg rather than the true value 1.0 kg — an 11% undershoot. Understanding this requires tracing through the closed-loop structure carefully.

The mass adaptive law (14) is driven by the composite error $\mathbf{s}_p$ via $\dot{\hat{m}} = \gamma_p \boldsymbol{\psi}_p^\top \mathbf{s}_p$. In steady-state (when $\mathbf{s}_p$ is small but not zero), the law seeks a value $\hat{m}$ that makes $\mathbf{s}_p \rightarrow \mathbf{0}$ in the translational error dynamics (12):

$$m\dot{\mathbf{s}}_p = m\boldsymbol{\psi}_p - \mathbf{f} - T(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i \quad (50)$$

Substituting (13): $m\boldsymbol{\psi}_p - k_v\mathbf{s}_p - \alpha_p\tilde{\mathbf{p}} - \hat{m}\boldsymbol{\psi}_p - T(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i = 0$, giving at equilibrium $(m - \hat{m})\boldsymbol{\psi}_p \approx k_v\mathbf{s}_p + T(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i$. When $\mathbf{B}$ is known (Part 2), $T(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i \rightarrow \mathbf{0}$ as $\mathbf{e}_R \rightarrow \mathbf{0}$, and $\mathbf{s}_p \rightarrow \mathbf{0}$ simultaneously, giving the correct equilibrium $\hat{m} \rightarrow m$.

However, in Part 5 with $\hat{\boldsymbol{\Gamma}} \neq \boldsymbol{\Gamma}$, the actual thrust delivered is $T_{\text{act}} = [\mathbf{B}\hat{\boldsymbol{\Gamma}}\mathbf{u}_{\text{virt}}]_1 = T_{\text{des}} + \Delta T$, where $\Delta T = [\mathbf{B}\tilde{\boldsymbol{\Gamma}}\mathbf{u}_{\text{virt}}]_1$. This constitutes a *persistent thrust bias*: even after attitude and position tracking have converged, $\Delta T \neq 0$ because $\tilde{\boldsymbol{\Gamma}} \neq \mathbf{0}$ (the allocation estimate has not converged). The bias enters the translational dynamics as an effective force offset. The mass adaptive law interprets this persistent offset as evidence that gravity is heavier than expected, and increases $\hat{m}$ to compensate. But wait — we observe an undershoot ($\hat{m} < m$), not an overshoot.

The sign of the bias is key. With $\hat{\boldsymbol{\Gamma}}(0) = 1.2\boldsymbol{\Gamma} > 0$, the over-estimated allocation delivers $\Delta T > 0$ (20% more thrust than commanded). The vehicle therefore rises faster than expected. The position controller responds by reducing $\mathbf{s}_p$ quickly, but the mass adaptive law (which integrates $\boldsymbol{\psi}_p \cdot \mathbf{s}_p$) sees a *smaller* time-integrated error than it would in the pure Part 2 scenario where no allocation bias exists. Since the law integrates a smaller error, it converges to a smaller steady-state $\hat{m}$. Mathematically: the allocation-induced ΔT partially compensates the gravity term in the dynamics, reducing the apparent weight of the vehicle to $m_{\text{apparent}} = m - \Delta T/g \approx 1.0 - 0.11 = 0.89$ kg, which is precisely what the mass adaptive law converges to.

This coupling between allocation and mass estimation is an inherent property of having two parallel adaptive loops estimating parameters that both enter the translational dynamics through the thrust channel. The σ -modification bounds the interaction but cannot decouple it without fundamentally restructuring the adaptive architecture. A possible

remedy (future work) is an orthogonal projection that separates the thrust-channel estimation from the mass estimation.

5.2.4 Parameter Estimation: Inertia

The inertia estimates $\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3$ remain near zero throughout, for the same PE reasons explained in Part 2. The additional perturbation $\Delta\boldsymbol{\tau}$ from the allocation mismatch does not significantly change this: $\Delta\boldsymbol{\tau}$ is small and does not add the high-frequency angular acceleration content needed to excite $\boldsymbol{\Psi}_o$. Attitude tracking quality is not impaired.

5.2.5 Allocation Matrix Estimate: Column-by-Column Analysis

Figure 8 shows all 16 elements of $\hat{\boldsymbol{\Gamma}}(t)$ converging toward $\boldsymbol{\Gamma}_{\text{true}}$. The Frobenius error reduces from 1.547 to 1.368 (11.6% reduction over 40 s). The convergence rate varies strongly by column; we now explain why.

Thrust column (col 0). The k -th row of $\hat{\boldsymbol{\Gamma}}$ is updated by $-\gamma_{\Gamma}[\mathbf{e}_{\text{adapt}}]_k[\mathbf{u}_{\text{virt}}]_0/(1 + \|\mathbf{u}_{\text{virt}}\|)$. The thrust channel $[\mathbf{u}_{\text{virt}}]_0 = T \approx 9.8 \text{ N}$ is persistently large throughout the simulation (the vehicle must continuously counteract gravity). The first element of $\mathbf{e}_{\text{adapt}}$ is $\mathbf{s}_p \cdot (\mathbf{R}\mathbf{z}_i)$, which is non-zero whenever position tracking is imperfect. Together, the product $[\mathbf{e}_{\text{adapt}}]_0 \cdot T$ provides sustained, large-magnitude driving of the thrust column, hence its fastest convergence. All four rows of col 0 decrease monotonically from 0.300 ($= 1.2 \times 0.250$) toward 0.250.

Roll/pitch columns (cols 1–2). The driving signal for col 1 is $\gamma_{\Gamma} \mathbf{e}_{\text{adapt}} \cdot \boldsymbol{\tau}_x$. The roll torque $\boldsymbol{\tau}_x$ is generated primarily by the differential in y -direction tracking, which along the smooth helical trajectory is small and slowly varying. The PE condition for cols 1–2 is therefore weakly satisfied, leading to slow convergence. This explains why $\hat{\Gamma}_{k1}$ and $\hat{\Gamma}_{k2}$ drift slowly toward their true values but do not approach them within 40 s.

Yaw column (col 3). The yaw channel $[\mathbf{u}_{\text{virt}}]_3 = \tau_z$ is driven by the sinusoidal yaw reference $\psi_d(t)$. Although τ_z is non-zero (up to $\approx -7 \text{ N m}$), its magnitude is much smaller than T , and its slowly-sinusoidal variation provides only modest PE for the yaw column. Furthermore, $s_{o,3}$ (the third component of $\mathbf{s}_o$ entering $\mathbf{e}_{\text{adapt}}$) is small once attitude tracking is established. This combination of small τ_z and small $s_{o,3}$ gives the yaw column the slowest update rate, consistent with what is visible in Figure 8: the col 3 elements change least over 40 s.

Full convergence would require sustained large τ_z commands (aggressive yaw manoeuvres) combined with persistent $\mathbf{s}_o$ excitation, which the smooth helical trajectory does not provide. This is the allocation-estimation analogue of the inertia PE problem in Part 2.

5.2.6 Motor Force Analysis

Initial transient. Figure 7 shows the motor forces under adaptive $\hat{\boldsymbol{\Gamma}}$. The initial transient is larger than Part 3 for a compounded reason: at $t = 0$, both the attitude correction

torque (due to 90° attitude error) and the 20% allocation over-estimate act simultaneously. As derived in Section 4.4, the attitude correction demands $\tau_z(0) \approx -7 \text{ N m}$. With $\hat{\Gamma}(0) = 1.2 \Gamma$, the commanded motor forces are:

$$\hat{\mathbf{F}}(0) = \hat{\Gamma}(0) \mathbf{u}_{\text{virt}}(0) = 1.2 \Gamma \mathbf{u}_{\text{virt}}(0) \quad (51)$$

The actual virtual inputs received are $\mathbf{u}_{\text{actual}} = \mathbf{B}\hat{\mathbf{F}} = 1.2 \mathbf{u}_{\text{virt}}$: 20% overshoot across all channels, meaning the vehicle simultaneously receives 20% more thrust and 20% more corrective torque than intended. The combination can drive some motors to large forces or deep into negative territory before the adaptive law corrects the allocation estimate.

Steady-state asymmetry. After $t \approx 5 \text{ s}$, motor forces stabilise at positive values, but the distribution is asymmetric: motors 1 and 2 produce $\approx 2.42 \text{ N}$ while motors 3 and 4 produce $\approx 2.41 \text{ N}$. This asymmetry is the fingerprint of the residual allocation error $\tilde{\Gamma} \neq \mathbf{0}$. To see why, consider that at hover the desired virtual input is $\mathbf{u}_{\text{virt}} = [mg, 0, 0, 0]^\top$. With $\hat{\Gamma} = \Gamma$ exactly, $\hat{\mathbf{F}} = \Gamma[mg, 0, 0, 0]^\top = [mg/4, mg/4, mg/4, mg/4]^\top$: symmetric. But with $\hat{\Gamma} \neq \Gamma$, the estimate introduces small off-diagonal contributions from the $\tilde{\Gamma}$ terms, leading to a non-uniform motor force distribution. The magnitude of this asymmetry is directly proportional to $\|\tilde{\Gamma}\|_F$, so it will decrease as the allocation estimate converges. The 0.01 N asymmetry observed (motors 3,4 producing 0.01 N less than motors 1,2) corresponds to the 11.6% remaining Frobenius error.

5.3 Comparison Summary

Table 3: Performance comparison across simulation parts

Metric	Part 2	Part 3	Part 5
Final $\ \tilde{\mathbf{p}}\ $ (m)	0.007	0.007	0.009
Final $\ \mathbf{e}_R\ $ (rad)	0.0016	0.0016	0.0035
Final $\hat{m}$ (kg)	1.0000	1.0000	0.893
Motor force distribution	N/A	Symmetric (2.42 N)	Asymmetric (2.41–2.42 N)
$\ \tilde{\Gamma}\ _F$	N/A	N/A	1.547 $\rightarrow$ 1.368 (11.6%)
Stability guarantee	Asymptotic	Asymptotic	UUB (σ -mod)
Known parameters	$m, \mathbf{J}$ unknown	$m, \mathbf{J}$ unknown	$m, \mathbf{J}, \mathbf{B}^{-1}$ unknown

The progression from Parts 2 to 5 demonstrates a clear trade-off: each additional unknown parameter reduces the strength of the stability guarantee (asymptotic $\rightarrow$ UUB) and slightly degrades tracking performance, while extending practical applicability to less well-characterised hardware. The degradation is modest ($\sim 30\%$ in final position error, $\sim 120\%$ in attitude error) relative to the additional 16-parameter estimation task added in Part 5, validating the robustness of the three-loop adaptive architecture.

A Attitude Regressor Derivation

For $\mathbf{J} = \text{diag}(\theta_1, \theta_2, \theta_3)$, expanding $\mathbf{J}\boldsymbol{\alpha} + \boldsymbol{\Omega} \times \mathbf{J}\boldsymbol{\Omega} = \boldsymbol{\tau}$ component-wise:

$$\begin{aligned}\tau_x &= \theta_1 \alpha_x + \Omega_y \theta_3 \Omega_z - \Omega_z \theta_2 \Omega_y \\ \tau_y &= \theta_2 \alpha_y + \Omega_z \theta_1 \Omega_x - \Omega_x \theta_3 \Omega_z \\ \tau_z &= \theta_3 \alpha_z + \Omega_x \theta_2 \Omega_y - \Omega_y \theta_1 \Omega_x\end{aligned}$$

The Jacobian $\partial\boldsymbol{\tau}/\partial[\theta_1, \theta_2, \theta_3]^\top$ gives (6) directly:

$$\boldsymbol{\Psi}_o = \begin{bmatrix} \alpha_x & -\Omega_z \Omega_y & \Omega_y \Omega_z \\ \Omega_z \Omega_x & \alpha_y & -\Omega_x \Omega_z \\ -\Omega_y \Omega_x & \Omega_x \Omega_y & \alpha_z \end{bmatrix} \quad (52)$$

B Sign Convention: Rotation Matrix vs. Quaternion

In Pliego-Jiménez [3], $\mathbf{s}_o = \boldsymbol{\omega}_r - \boldsymbol{\omega}$ (desired minus actual), so $\boldsymbol{\tau} = k_\omega \mathbf{s}_o + \alpha_o \tilde{\boldsymbol{\epsilon}} + \boldsymbol{\Psi}_o \hat{\boldsymbol{\theta}}$ with *positive* gains gives negative feedback: when $\boldsymbol{\omega} < \boldsymbol{\omega}_r$ ($\mathbf{s}_o > 0$), torque is positive and increases $\boldsymbol{\omega}$. This is correct.

In our rotation-matrix formulation, $\mathbf{e}_\Omega = \boldsymbol{\Omega} - \mathbf{R}^\top \mathbf{R}_d \boldsymbol{\Omega}_d$ (actual minus desired). Using positive gains here creates *positive feedback* (when $\boldsymbol{\Omega} > \boldsymbol{\Omega}_r$, torque increases, driving $\boldsymbol{\Omega}$ further away), leading to divergence. The correct law is (18) with negative gains. This sign error was the root cause of all early simulation failures. Once corrected, the simulation converged reliably.

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